Exercises

Exercise 6.1 · Euclidean divisions.

1. Carry out the Euclidean division of 2,026 by 17.
2. Carry out the Euclidean division of -47 by 5, then of -35 by 4.
3. Determine the natural numbers n whose quotient in the division by 6 equals the remainder.

Exercise 6.2 · Remainders of powers.

1. Determine the remainder of 3^{100} in the division by 5.
2. Determine the remainder of $5^{2026} + 2^{2026}$ in the division by 3.

Exercise 6.3 · Sums of two squares.

1. Prove that for every integer x, $x^2 \equiv 0$ or $x^2 \equiv 1 \pmod 4$.
2. Deduce that an integer congruent to 3 modulo 4 is not the sum of two squares of integers.
3. Can the integer 2,027 be written $x^2 + y^2$ with x, y integers?

Exercise 6.4 · Divisibility test for 11.

1. Prove that $10 \equiv -1 \pmod{11}$, then that $10^k \equiv (-1)^k \pmod{11}$.
2. Deduce that an integer is divisible by 11 if and only if the alternating sum of its digits, starting from the units, $a_0 - a_1 + a_2 - \dots$, is.
3. Is the integer 918,082 divisible by 11?

Exercise 6.5 · Divisibility for every integer. Prove that for every natural number n:

1. $4^n - 1$ is divisible by 3;
2. $n^3 - n$ is divisible by 6.

Exercise 6.6 · Congruence equations.

1. Solve $3x \equiv 5 \pmod 7$. First look for an integer u such that $3u \equiv 1 \pmod 7$.
2. Solve $x^2 \equiv 2 \pmod 7$ with a table.

Exercise 6.7 · The last two digits. Determine the last two digits of the decimal expansion of 7^{2026}.